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Section : A

RollNo. : 201500340(29)

Lab Assignment -3

❖ *Creating Table 1- Student ->*

```
CREATE TABLE Student(sID INTEGER, sName VARCHAR(10), GPA NUMERIC(2,1),
sizeHS INTEGER,DoB DATE);
```

```
INSERT INTO student VALUES
(123,"Amy",3.9,1000,'1996-06-26'),
(234,'Bob',3.6,1500,'1995-04-07'),
(345, 'Craig', 3.5, 500, '1995-02-04'),
(456, 'Doris', 3.9, 1000,'1997-07-24'),
(567, 'Edward', 2.9, 2000,'1996-12-21'),
(678, 'Fay', 3.8, 200 ,'1996-08-27'),
(789, 'Gary', 3.4, 800 ,'1996-10-08'),
(987, 'Helen', 3.7, 800 ,'1997-03-27'),
(876, 'Irene', 3.9, 400 ,'1996-03-07'),
(765, 'Jay', 2.9, 1500,'1998-08-08'),
(654, 'Amy', 3.9, 1000,'1996-05-26'),
(543, 'Craig', 3.4, 2000,'1998-08-27');
```

```
SELECT * from Student;
```

sID	sName	GPA	sizeHS	DoB
123	Amy	3.9	1000	1996-06-26
234	Bob	3.6	1500	1995-04-07
345	Craig	3.5	500	1995-02-04
456	Doris	3.9	1000	1997-07-24
567	Edward	2.9	2000	1996-12-21
678	Fay	3.8	200	1996-08-27
789	Gary	3.4	800	1996-10-08
987	Helen	3.7	800	1997-03-27
876	Irene	3.9	400	1996-03-07
765	Jay	2.9	1500	1998-08-08
654	Amy	3.9	1000	1996-05-26
543	Craig	3.4	2000	1998-08-27

❖ ***Creating Table 2- College ->***

```
CREATE TABLE College(collegeName VARCHAR(10), state VARCHAR(10),
enrolment INTEGER);
```

```
Insert into college values
('Stanford' ,'CA' ,15000),
('Berkeley', 'CA', 36000),
('MIT', 'MA', 10000),
('Cornell', 'NY', 21000),
('Harvard', 'MA', 50040);
```

```
SELECT * from College;
```

cName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

❖ ***Creating Table 3- Apply ->***

```
CREATE TABLE Apply(sID INTEGER, cName VARCHAR(10), major VARCHAR(20),
decision char(1));
```

```
INSERT INTO apply VALUES
(123, 'Stanford', 'CS', 'Y'),
(123, 'Stanford', 'EE', 'N'),
(123, 'Berkeley', 'CS', 'Y'),
(123, 'Cornell', 'EE', 'Y'),
```

```
(234, 'Berkeley', 'biology', 'N'),
(345, 'MIT', 'bioengineering', 'Y'),
(345, 'Cornell', 'bioengineering', 'N'),
(345, 'Cornell', 'CS', 'Y'),
(345, 'Cornell', 'EE', 'N'),
(678, 'Stanford', 'history', 'Y'),
(987, 'Stanford', 'CS', 'Y'),
(987, 'Berkeley', 'CS', 'Y'),
(876, 'Stanford', 'CS', 'N'),
(876, 'MIT', 'biology', 'Y'),
(876, 'MIT', 'marine biology', 'N'),
(765, 'Stanford', 'history', 'Y'),
(765, 'Cornell', 'history', 'N'),
(765, 'Cornell', 'psychology', 'Y'),
(543, 'MIT', 'CS', 'N');
```

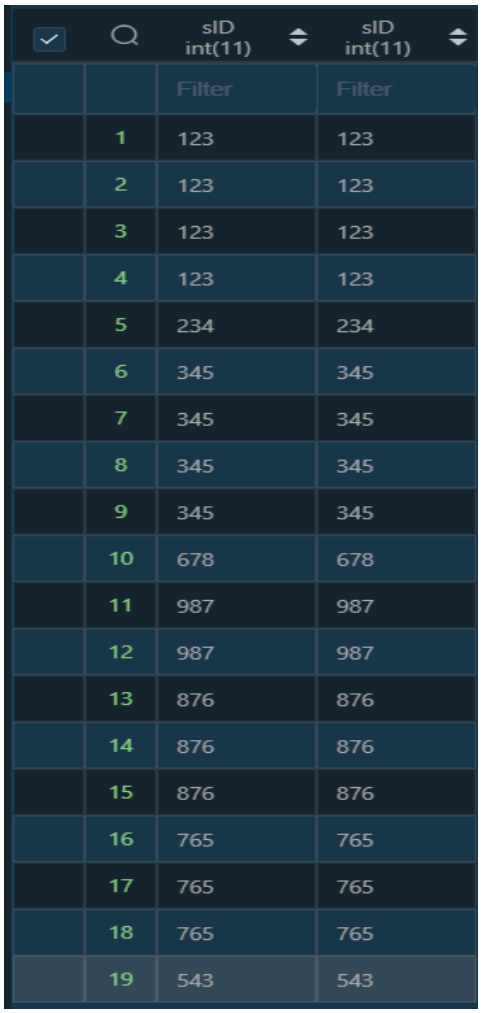
```
SELECT * from Apply;
```

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

Queries

1. **Produce a combine table in which each student is combine with every otherapplication.**

```
SELECT Student.sID,Apply.sID FROM Student
INNER JOIN Apply
ON Student.sID = Apply.sID;
```



	sID int(11)	sID int(11)
	Filter	Filter
1	123	123
2	123	123
3	123	123
4	123	123
5	234	234
6	345	345
7	345	345
8	345	345
9	345	345
10	678	678
11	987	987
12	987	987
13	876	876
14	876	876
15	876	876
16	765	765
17	765	765
18	765	765
19	543	543

2. **Give Student ID, name, GPA and name of college and major each studentapplied to.**

```
SELECT
Student.sID,Student.sName,Student.GPA,Apply.cName,Apply.major FROM
Student INNER JOIN Apply ON Student.sID = Apply.sID;
```

		sID int(11)	sName varchar(10)	GPA decimal(2,1)	cName	major
		Filter	Filter	Filter	Filter	Filter
	1	123	Amy	3.9	Stanford	CS
	2	123	Amy	3.9	Stanford	EE
	3	123	Amy	3.9	Berkeley	CS
	4	123	Amy	3.9	Cornell	EE
	5	234	Bob	3.6	Berkeley	biology
	6	345	Craig	3.5	MIT	bioengineering
	7	345	Craig	3.5	Cornell	bioengineering
	8	345	Craig	3.5	Cornell	CS
	9	345	Craig	3.5	Cornell	EE
	10	678	Fay	3.8	Stanford	history
	11	987	Helen	3.7	Stanford	CS
	12	987	Helen	3.7	Berkeley	CS
	13	876	Irene	3.9	Stanford	CS
	14	876	Irene	3.9	MIT	biology
	15	876	Irene	3.9	MIT	marine biology
	16	765	Jay	2.9	Stanford	history
	17	765	Jay	2.9	Cornell	history
	18	765	Jay	2.9	Cornell	psychology
	19	543	Craig	3.4	MIT	CS

3. Find detail of applications who applied to California State.

```
SELECT Apply.cName,College.cName,College.state FROM Apply
INNER JOIN College
On Apply.cName=College.cName And College.state='CA';
```

		cName varchar(10)	cName varchar(10)	state
		Filter	Filter	Filter
	1	Stanford	Stanford	CA
	2	Stanford	Stanford	CA
	3	Berkeley	Berkeley	CA
	4	Berkeley	Berkeley	CA
	5	Stanford	Stanford	CA
	6	Stanford	Stanford	CA
	7	Berkeley	Berkeley	CA
	8	Stanford	Stanford	CA
	9	Stanford	Stanford	CA

4. **IDs, name, GPA of students and name of college with GPA > 3.7 applying toStanford.**

```
SELECT Student.sID,Student.sName,Student.GPA,Apply.sID,Apply.cName
FROM Student
INNER JOIN Apply On Student.sID = Apply.sID
And Student.GPA > 3.7 And Apply.cName='Stanford';
```

		sID int(11)	sName varchar(10)	GPA decimal(2,1)	sID int(11)	cName
		Filter	Filter	Filter	Filter	Filter
	1	123	Amy	3.9	123	Stanford
	2	123	Amy	3.9	123	Stanford
	3	678	Fay	3.8	678	Stanford
	4	876	Irene	3.9	876	Stanford

5. **Find detail of Student who apply to CS major and their application are rejected.**

```
SELECT Student.sID,Student.sName,Apply.sID,Apply.major,Apply.decision
FROM Student INNER JOIN Apply On Student.sID = Apply.sID And
Apply.major='CS' And Apply.decision='N';
```

	sID int(11)	sName varchar(10)	sID int(11)	major	decision
	Filter	Filter	Filter	Filter	Filter
1	876	Irene	876	CS	N
2	543	Craig	543	CS	N

6. **Find detail of student and application who applied to colleges at New York.**

```
SELECT Apply.cName,Apply.major,College.cName,College.enrollment,
College.state FROM Apply INNER JOIN College On Apply.cName =
College.cName And College.state='NY';
```

	cName varchar(10)	major varchar(20)	cName varchar(10)	enrollment	state
	Filter	Filter	Filter	Filter	Filter
1	Cornell	EE	Cornell	21000	NY
2	Cornell	bioengineering	Cornell	21000	NY
3	Cornell	CS	Cornell	21000	NY
4	Cornell	EE	Cornell	21000	NY
5	Cornell	history	Cornell	21000	NY
6	Cornell	psychology	Cornell	21000	NY

7. **Find detail of student who have not applied to any of college.**

```
SELECT Student.sID,Student.sName,Student.GPA,Student.sizeHs
FROM Student LEFT JOIN Apply ON Student.sID = Apply.sID
WHERE Apply.sID IS NULL;
```

	sID int(11)	sName varchar(10)	GPA decimal(2,1)	sizeHs
	Filter	Filter	Filter	Filter
1	456	Doris	3.9	1000
2	567	Edward	2.9	2000
3	789	Gary	3.4	800
4	654	Amy	3.9	1000

8. **Find college where no student have applied.**

```
SELECT cName FROM College Left JOIN Apply
ON College.cname = Apply.cName WHERE Apply.cName IS NULL
```

	collegeName varchar(10)
	Filter
1	Harvard

9. **Find sID who have only one application.**

```
SELECT sID FROM Apply GROUP BY sID
HAVING count(sID) = 1;
```

	sID int(11)
	Filter
1	234
2	543
3	678

10. **Find name and GPA of applicants who apply to any college whoseenrollment is not more than 25000.**

```
SELECTStudent.sName,Student.GPA,Student.sID,Apply.sID,Apply.cName,
College.collegeName,College.enrollment FROM ((Student INNER JOIN Apply
ON Student.sID = Apply.sID) INNER JOIN College ON
Apply.cName = College.collegeName And College.enrollment < 25000 );
```

	sName varchar(10)	GPA decimal(2,1)	sID int(11)	sID int(11)	cName	collegeName	enrollment
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	Amy	3.9	123	123	Stanford	Stanford	15000
2	Amy	3.9	123	123	Stanford	Stanford	15000
3	Amy	3.9	123	123	Cornell	Cornell	21000
4	Craig	3.5	345	345	MIT	MIT	10000
5	Craig	3.5	345	345	Cornell	Cornell	21000
6	Craig	3.5	345	345	Cornell	Cornell	21000
7	Craig	3.5	345	345	Cornell	Cornell	21000
8	Fay	3.8	678	678	Stanford	Stanford	15000
9	Helen	3.7	987	987	Stanford	Stanford	15000
10	Irene	3.9	876	876	Stanford	Stanford	15000
11	Irene	3.9	876	876	MIT	MIT	10000
12	Irene	3.9	876	876	MIT	MIT	10000
13	Jay	2.9	765	765	Stanford	Stanford	15000
14	Jay	2.9	765	765	Cornell	Cornell	21000
15	Jay	2.9	765	765	Cornell	Cornell	21000
16	Craig	3.4	543	543	MIT	MIT	10000

11.Find pair of students (sID) having same GPA. (each pair should occur justonce in result).

```
Select student.sID, x.sID,student.GPA
from Student, Student x where student.sID < x.sID and student.GPA =
x.GPA;
```

	sID int(11)	sID int(11)	GPA decimal(2,1)
	Filter	Filter	Filter
1	456	456	3.9
2	789	789	3.4
3	876	876	3.9
4	876	876	3.9
5	876	876	3.9
6	765	765	2.9
7	654	654	3.9
8	654	654	3.9